21 November 1973
SUPERSEDING
MIL-F-45534(WC)
4 February 1963
(see section 6)

MILITARY SPECIFICATION

FIRING BOX ASSEMBLY

1. SCOPE

1.1 This specification covers one type of firing box assembly used to energize the firing circuit on the M91 Rocket Launcher.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

FSC 1055

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DRAWINGS

U. S. Army Weapons Command		
F8425720	-	Firing Box Assembly.
IEL8425720	-	Inspection Equipment List.

(Product drawings referenced in this specification form a part of F8425720: inspection equipment drawings form a part of IEL8425720.)

PUBLICATIONS

U. S. Army Weapons Command		
P8425720	-	Packaging Data Sheet for Firing Box Assembly.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include a pilot pack (see 5.1).

* 3.2 Materials, construction and design. Firing box assemblies shall conform to the materials, construction, and design requirements specified herein and on Drawing F8425720.

3.2.1 Threaded fasteners. Fasteners shall be torqued at assembly to provide a tight fit of the attached parts. Torque requirements are specified as follows:

Thread size	Torque
No. 4 - 40	3.5 - 4.5 inch lb.
No. 6 - 32	7 - 9 inch lb.
No. 8 - 32	14 - 17 inch lb.
No. 10 - 24	20 - 25 inch lb.

* 3.2.2 Wiring. Wires and leads shall be free of kinks and shall not project into areas of moving parts. Wires may be taped or tied at the option of the contractor.

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* 3.2.3 Cover centering and loading. The cover shall be centered over the box assembly and loaded to provide 1/64 + 1/32 compression of the seal when the cover is closed and latched.

3.3 Performance requirements.

3.3.1 Electrical circuit. The electrical circuit shall conform to the wiring diagram, Drawing F8425768 and show no evidence of shorts between circuits or to the cover.

* 3.3.2 Battery activating mechanism. With the slotted lever in the position shown on Drawing F8425768 and the knob of the handle held in contact with the top of the slotted lever, movement of the handle from left to right shall activate the mechanism as follows:

a. The spring loaded lever, Drawing B8425682, shall be pushed to the right causing the pin end of the plunger, Drawing B8425770, to recede to a maximum protrusion of .001 inch above the bottom surface of the battery well.
b. When the pin in the handle overrides the spring loaded lever and releases it, the pin end of the plunger shall be driven to protrude not less than 9/64 inch above the bottom surface of the battery well.
c. The handle and slotted lever shall be returned to their original position by withdrawing the handle to the fullest extent within the slotted lever and moving the handle from right to left. The pin in the handle shall pass above the spring loaded lever without making contact.

3.4 Marking. Firing box assemblies shall be marked in accordance with the applicable drawings and MIL-STD-130.

* 3.5 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burs, deformations or missing operations which may affect serviceability, functioning, operation, appearance or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with contract requirements (see 6.1). The first article shall be representative of the production processes used during quantity production. The first article shall be subjected to quality conformance inspection specified herein, and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lots. Unless otherwise specified, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions and the contractor's delivery schedule. Firing box assemblies shall be assembled from component parts that have met all inspection requirements specified herein.

4.5.2 Examination.

* 4.5.2.1 Component parts and concurrent repair parts. Examination of component parts shall be performed in accordance with the criteria specified in the contract (see 6.1). Examination of these parts shall be accomplished prior to their assembly into the end items or submission for acceptance as repair parts.

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4.5.2.2 Firing box assemblies. Examination of firing box assemblies shall be as specified herein.

* 4.5.2.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each group of characteristics.
b. Where "Visual" is specified as the inspection method for dimensional inspection the characteristic shall be either scaled or compared with a specimen of known acceptable quality that has been established as an inspection standard for conformance to the requirements of the drawing.
c. Where "Visual" is specified as the inspection method for protective coating, the coating shall be visually examined for completeness, uniformity in appearance and color, and for freedom from pits, corrosion, scratches, and worn or bare spots.
d. Where "Visual" is specified as the inspection method for functioning assembly requirements, the assembly shall be visually examined for completeness and manually operated for functioning requirements.
e. Where "SMTE" (Standard Measuring and Test Equipment) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection and/or testing equipment approved by the Government.
f. Examination for workmanship shall include examination for defects as specified in 3.5. Assemblies shall be compared with a speciman of known acceptable quality.

4.5.2.2.1.1 Firing box assembly (Drawing F8425720).

	Categories	Defect	Inspection method
	Critical:	None defined.	
	Major:	AQL = 1.5 percent	
	101	Any missing or type part not as specified.	Visual
	102	Any part improperly assembled or attached.	Visual
*	103	Threaded fasteners improperly torqued (see 3.2.1).	SMTE
*	104	Improper wiring (see 3.2.2).	Visual

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Categories	Defect	Inspection method
105	Cover not centered and loaded to provide compression of rubber seal (see 3.2.3).	SMTE
106	Rubber seal not adherent, puckered, or incorrectly spliced.	Visual
107	Hardware including catches, handle and hinges do not operate as intended.	Visual
108	Loose or improperly headed rivets.	Visual
Minor:	AQL = 4.0 percent	
201	Painting defects including discontinuities, sags, runs, blisters or wrong color.	Visual
202	Identification marking incorrect, mislocated, improper size or wrong color (see 3.4).	Visual
* 203	Workmanship (see 3.5).	Visual

4.5.3 Testing.

* 4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete firing box assembly. If test requirements cited herein are not met, acceptance of the firing box assembly shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical, and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defects in future production.
c. Examine firing box assemblies, partially assembled firing box assemblies and components (including components and subassemblies at in-process or final assembly) to assure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned firing box assembly for retest.

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* 4.5.3.2 Electrical circuit testing. Each firing box assembly in the inspection lot shall be tested for electrical circuit conformance (see 3.3.1) using the test method specified in 4.6.1. Failure of any firing box assembly to meet these requirements shall be cause for rejection and the contractor shall perform the corrective action specified in 4.5.3.1.

* 4.5.3.3 Battery activating mechanism testing. Each firing box assembly in the inspection lot shall be tested for conformance of the battery activating mechanism using the test method specified in 4.6.2. Failure of any firing box assembly to meet this requirement shall be cause for rejection and the contractor shall perform the corrective action specified in 4.5.3.1.

* 4.5.3.4 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be as specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. These tests shall be accomplished prior to assembly of parts into the end item.

4.5.3.5 Packaging examination and testing. Unless otherwise specified (see 6.1), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.4 Inspection equipment.

* 4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, (see 6.1), responsibility for acquisition, maintenance and disposition of acceptance and test equipment prescribed on the List of Inspection Equipment IEL8425720 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

* 4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

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4.6 Test methods.

4.6.1 Electrical circuit test. Using an ohmmeter or other suitable instrument approved by the Government, the electrical circuit shall be tested for conformance to Drawing F8425768 as follows:

a. Install safety plug, Drawing B8425680, in mating receptacle and remove receptacle caps from each side of firing box.
b. With the toggle switch in each of its three positions, continuity of circuits shall exist from pins C and D of connector MS3102R22-22P to the minus (-) terminal of the battery housing and to pin sockets C and D of connector MS3102R20-22S.
c. With the toggle switch in "fire" position, continuity of circuits shall exist from pins A and B of connector MS3102R22-22P to the plus (+) terminal of battery housing and to pin sockets A, B, E and F of connector MS3102R20-22S.
d. With the toggle switch in "reset" position, continuity of circuits shall exist from pins A and B of connector MS3102R22-22P to the plus (+) terminal of the battery housing and to pin sockets E and F of connector MS3102R20-22S.
e. Remove the safety plug and repeat b, c, and d above. Open circuits must exist at all pins of connector MS3102R20-22S.
f. Check to assure that toggle switch automatically returns to the "off" position when released from the "fire" and "reset" positions.

4.6.2 Battery activating mechanism test. The activating mechanism shall be tested by operating the handle as described in 3.3.2. Parts shall be examined for freedom of movement and travel of plunger.

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. The pilot pack shall consist of a complete firing box assembly packaged in accordance with Packaging Data Sheet P8425720 for the level of protection specified in the contract (see 6.1), packed level C and forwarded in accordance with 3.1.

* 5.2 Preservation, packaging, packing and marking. The firing box assembly shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P8425720 for the level of protection specified in the contract (see 6.1).

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* 5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or the requirements specified, and for the level of protection specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the firing box assembly showing applicable revision dates.
c. Requirement and number of firing box assemblies to be submitted for first article by the contractor (see 3.1, 4.4 and 5.1).
d. Examination and testing criteria for component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.4).
e. Packaging examination and testing, if different (see 4.5.3.5).
f. List of inspection equipment, responsibilities for acquisition, maintenance, and disposition of, if other than as specified (see 4.5.4.1).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
h. Packaging instructions for repair parts (see 5.3).

* 6.2 Management control systems requirement. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

* 6.3 Unless otherwise specified (see 6.1 f), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

* 6.4 Supersession data. The basic issue of MIL-F-45534(WC) dated 4 February 1963 included the requirements of Rock Island Arsenal Purchase Description, RIAPD-637 dated 25 May 1961.

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* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last issue.

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